Feature Dictionary

Customer Personality Segmentation | Module 5: Feature Engineering & Data Preprocessing (CPS-M05)

This document lists every feature I engineered from the raw customer dataset, what it's calculated from, why I built it, and what it's expected to contribute once we get to clustering. Reference date used for all age/tenure calculations is 2014-06-30 (one day after the latest enrollment date in the dataset).

Feature	Formula / Calculation	Purpose	Business Significance	Expected Impact
Customer_Age	Reference_Year - Year_Birth	Turns a raw birth year into an interpretable age.	Age drives spending capacity, life stage, and product preferences (e.g. families vs. retirees).	Expected to be one of the stronger separators between clusters.
Customer_Tenure_Days	Reference_Date - Dt_Customer	Measures how long someone has been a customer, in days.	Distinguishes new sign-ups from long-term loyal customers, who often behave very differently.	Useful for a 'loyalty' axis alongside spending.
Total_Children	Kidhome + Teenhome	Combines the two separate child-count columns into one.	Households with children buy differently (more staples, less discretionary/luxury spend).	Expected to correlate with lower luxury spend, higher deal usage.
Family_Size	1 + Is_Partnered + Total_Children	Estimates total household size from marital status and children.	Bigger households mean different purchase volume and category needs than singles.	Helps separate 'family shopper' segments from single/couple segments.
Total_Spending	Sum of MntWines, MntFruits, MntMeatProducts, MntFishProducts, MntSweetProducts, MntGoldProds	Single number for how much a customer spends across all product categories.	The most direct measure of a customer's overall value to the business.	Expected to be the primary driver separating high-value from low-value segments.
Total_Purchases	Sum of NumDealsPurchases, NumWebPurchases, NumCatalogPurchases, NumStorePurchases	Single number for total purchase frequency across all channels.	Frequency and spend don't always move together — this captures the frequency side separately.	Combined with Total_Spending, helps separate 'frequent small buyers' from 'occasional big spenders'.
Total_Campaign_Acceptance	Sum of AcceptedCmp1-5 + Response	Counts how many of the six marketing campaigns a customer said yes to.	Direct measure of marketing responsiveness / persuadability.	Expected to identify a distinct 'campaign-responsive' segment worth targeting first.
Avg_Spending_Per_Purchase	Total_Spending / Total_Purchases (0 if no purchases)	Average amount spent per individual purchase — a proxy for basket size.	Two customers can have the same Total_Spending with very different shopping styles (many small trips vs. few big ones).	Adds texture beyond totals; helps separate 'bulk buyers' from 'frequent small buyers'.
Digital_Engagement	NumWebPurchases + NumWebVisitsMonth	Combines actual web purchases with monthly web visits into one online-activity score.	Captures both browsing and buying behavior online, not just conversions.	Expected to separate digitally-engaged customers from in-store-only shoppers.

Feature	Formula / Calculation	Purpose	Business Significance	Expected Impact
Deal_Dependency	NumDealsPurchases / Total_Purchases (0 if no purchases)	The share of a customer's purchases that were deal/discount driven.	Flags price-sensitive customers who may need different offers than full-price shoppers.	Expected to be a key differentiator between 'bargain hunter' and 'brand loyal' segments.
Preferred_Shopping_Channel	Channel (Web/Catalog/Store/Deals) with the highest purchase count	Identifies which single channel a customer relies on most.	Channel behavior has direct implications for where to focus marketing spend per segment.	One-hot encoded for clustering; expected to create channel-based sub-segments.
Product_Preference	Product category (Wines/Fruits/Meat/Fish/Sweets/Gold) with the highest spend	Identifies which single product category a customer favors.	Useful for category-specific promotions and cross-sell targeting.	One-hot encoded for clustering; expected to reveal category-loyal segments (e.g. wine enthusiasts).
Customer_Activity_Level	Recency binned: 0-30 days = High, 31-60 = Medium, 61-100 = Low	Converts the continuous Recency column into three interpretable activity tiers.	Easier to act on 'High/Medium/Low activity' in a marketing plan than a raw day count.	Label-encoded (ordinal); expected to help flag at-risk/lapsing customers.

Notes on Encoding & Transformation Applied to These Features

Preferred_Shopping_Channel and Product_Preference are nominal categories, so they're one-hot encoded rather than label encoded — there's no natural order between 'Web' and 'Store', for example. Customer_Activity_Level is ordinal (Low < Medium < High) so it's label encoded to preserve that order as a single numeric column. Several of the numeric features above (Income, Total_Spending, Avg_Spending_Per_Purchase, and the individual Mnt* columns) were right-skewed and received a log1p transform before scaling — see the Feature Selection Report and the notebook for the full skewness analysis.